Michael Asemota

Arlington, TX 76010 | (469) 767 0976 | michael.asemotaa@gmail.com
GitHub Portfolio: https://ybmikey.github.io/Michael-A-Projects/index.html
Candidate for Engineering, seeking internships for 2026 & 2027

Education

University of Texas at Arlington
2023 - Present (Expected 2027)
- Industrial Engineering Major

Allen High School
2019 - 2023
- 3.8 Weighted GPA

Technical Skills
- CAD: SolidWorks, AutoCAD, Onshape, Fusion360, MATLAB
- Programming: C++, Python, Arduino
- Tools/Systems: Arduino Uno, Robotics, 3D Printing, Gimbal Systems
- Soft Skills: Six Sigma - White Belt, Lean Manufacturing, MS Office Applications (Excel, Word, etc.), DFM, DFA, FMEA, Fixture Design, Root Cause Analysis, Bottleneck Analysis

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Job Experience

Verlo (Technical Assistant)
2026 - Present
- Matched purchase requests to optimized mattress configurations using parameters such as foam density, and coil system architecture.
- Used spreadsheets to organize and analyze product data, improving collection workflows and efficiency.
- Leveraged customer and product data to identify process improvements in manufacturing and distribution efficiency.

Skechers (Associate Manager)
2022 - 2024
- Supervised daily operations for a high-volume retail store, leading 10–15 associates and increasing sales performance by 15–20% through structured coaching and performance tracking.
- Reduced stockouts by 25% by monitoring inventory and coordinating restocking using MS Office tools.
- Improved labor efficiency by 10–15% through sales data analysis and optimized scheduling, while implementing stockroom process improvements that reduced retrieval time by 30%.

Activities

NSBE (National Society of Black Engineers)
2023 - Present
- Participated in technical workshops and engineering events promoting diversity in STEM
- Engaged in collaborative engineering projects and networking opportunities with industry professionals
- Contributed to community outreach and professional development initiatives

Formula SAE
2023 - Present
- Contributed to the design, fabrication, and testing of high-performance vehicles in Formula SAE
- Applied knowledge in aerodynamics, mechanical systems, and electronics to improve vehicle performance
- Collaborated across sub-teams to troubleshoot design issues and optimize system integration

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Leadership

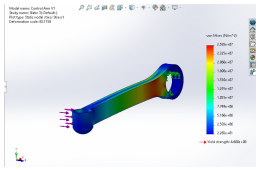
Robomasters (Gimbal Team Lead - 3D Modeling & Secretary)
2024 - Present
- Organized and led structured team meetings, managed technical documentation, and streamlined communication workflows to support cross-functional coordination.
- Served as liaison between internal teams and external stakeholders, including sponsors and competition organizers, ensuring clear requirements alignment and project tracking.
- Directed end-to-end gimbal system development, overseeing mechanical assembly, integration, and performance optimization.
- Guided system testing and calibration efforts, improving control accuracy and operational stability through iterative validation.
- Designed and assembled a precision gimbal mechanism incorporating high-torque motors and structural components, collaborating with teammates to refine control algorithms and ensure smooth, responsive motion.

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Suspension Control Arm | CAD & FEA (In Progress)



What

- Modeled and refined a Mercedes-Benz S550 control arm in **SolidWorks**, using static FEA to evaluate stress concentrations.
- Revised width, fillets, and ball-joint geometry; documented material research and design iterations.

Why

- Explore how suspension geometry, material selection, and load paths affect structural performance.

Results

- Measured **17.6% lower average local von Mises stress** in V2 (341.21 to 281.15 MPa) across five probe locations near the bushing. Preliminary simulation result; mesh and boundary-condition verification ongoing.

RoboMasters Standard Robot



What

- Contributed to chassis, drivetrain, and dual-axis gimbal design; integrated mechanical interfaces with electrical and controls teams.
- Refined components in **SolidWorks and Fusion 360** for manufacturing, assembly, and serviceability.

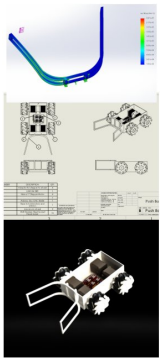
Why

- Develop a competition-ready robot with stable mobility and accurate targeting during motion.

Results

- Delivered a functional robot that competed in the **RoboMasters 1v1 North America tournament**, with responsive targeting and a modular design for continued iteration.

Arduino Mecanum Robot



What

- Built an omnidirectional platform; modeled mechanical systems in **SolidWorks and Onshape** for wheel alignment and load distribution.
- Programmed **C++ on Arduino Uno R4** for acceleration, directional control, and repeatable object manipulation.

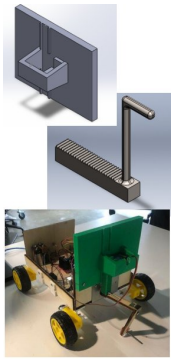
Why

- Achieve precise motion in a constrained workspace while addressing mechanical and control challenges.

Results

- Completed the task in **under 30 seconds**, demonstrating accurate positioning, reliable control, and consistent system performance.

Arduino Based Robot



What

- Designed an Arduino-controlled **rack-and-pinion** mechanism for precision object transfer.
- Modeled parts in **SolidWorks** for alignment and tolerance control; implemented **C++** actuator timing and sequencing.

Why

- Perform a repeatable transfer within spatial constraints without crossing a defined boundary.

Results

- Completed the objective in **under 30 seconds with no faults**, demonstrating dependable mechanical design and repeatable control.